

```
from google.colab import files
```

```
uploaded = files.upload()
```

Choose files 2 files
gateway_export.csv(text/csv) - 30630 bytes, last modified: 24/08/2026 - 100% done
ledger.csv(text/csv) - 31635 bytes, last modified: 24/08/2026 - 100% done
 Saving gateway_export.csv to gateway_export.csv
 Saving ledger.csv to ledger.csv

```
import pandas as pd
```

```
ledger_df = pd.read_csv("ledger.csv")
gateway_df = pd.read_csv("gateway_export.csv")
```

```
print("Ledger rows:", len(ledger_df))
print("Gateway rows:", len(gateway_df))
```

```
print("\nLedger columns:")
print(ledger_df.columns.tolist())
```

```
print("\nGateway columns:")
print(gateway_df.columns.tolist())
```

```
Ledger rows: 547
Gateway rows: 530
```

```
Ledger columns:
['transaction_id', 'user_id', 'merchant_id', 'transaction_time', 'amount_inr', 'payment_method']
```

```
Gateway columns:
['transaction_id', 'user_id', 'merchant_id', 'transaction_time', 'amount_inr', 'payment_method']
```

```
def reconcile_payments(ledger_df, gateway_df):
    # Make copies so the original DataFrames are not changed
    ledger = ledger_df.copy()
    gateway = gateway_df.copy()

    # Make sure transaction IDs are comparable
    ledger_ids = set(ledger["transaction_id"])
    gateway_ids = set(gateway["transaction_id"])

    # 1. Transactions present in ledger but missing from gateway
    missing_in_gateway_ids = ledger_ids - gateway_ids

    missing_in_gateway = ledger[
        ledger["transaction_id"].isin(missing_in_gateway_ids)
    ].copy()

    # 2. Transactions present in gateway but missing from ledger
    missing_in_ledger_ids = gateway_ids - ledger_ids

    missing_in_ledger = gateway[
        gateway["transaction_id"].isin(missing_in_ledger_ids)
    ].copy()

    # 3 & 4. Compare matching transactions using merge
    matched = pd.merge(
        ledger,
        gateway,
        on="transaction_id",
```

```

        how="inner",
        suffixes=("_ledger", "_gateway")
    )

    # 3. Amount mismatches
    matched["amount_difference"] = (
        matched["amount_inr_ledger"] -
        matched["amount_inr_gateway"]
    )

    amount_mismatches = matched[
        matched["amount_difference"] != 0
    ].copy()

    # 4. Status mismatches
    status_mismatches = matched[
        matched["status_ledger"] != matched["status_gateway"]
    ].copy()

    return (
        missing_in_gateway,
        missing_in_ledger,
        amount_mismatches,
        status_mismatches
    )

```

```

missing_in_gateway, missing_in_ledger, amount_mismatches, status_mismatches = reconcile_pay
    ledger_df,
    gateway_df
)

```

```

print("Transactions missing in gateway:", len(missing_in_gateway))
print("Transactions missing in ledger:", len(missing_in_ledger))
print("Amount mismatches:", len(amount_mismatches))
print("Status mismatches:", len(status_mismatches))

```

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Transactions missing in gateway: 27
Transactions missing in ledger: 10
Amount mismatches: 16
Status mismatches: 9

```

```

print("=== TRANSACTIONS MISSING IN GATEWAY ===")
display(missing_in_gateway)

print("=== TRANSACTIONS MISSING IN LEDGER ===")
display(missing_in_ledger)

print("=== AMOUNT MISMATCHES ===")
display(
    amount_mismatches[
        [
            "transaction_id",
            "amount_inr_ledger",
            "amount_inr_gateway",
            "amount_difference"
        ]
    ]
)

print("=== STATUS MISMATCHES ===")
display(
    status_mismatches[
        [

```

```
        "transaction_id",  
        "status_ledger",  
        "status_gateway"  
    ]  
]  
)
```



```
=== TRANSACTIONS MISSING IN GATEWAY ===
```

	transaction_id	user_id	merchant_id	transaction_time	amount_inr	payment_method	
0	TXN100000	350	16	2026-01-29 11:13:00	2999	Wallet	
5	TXN100005	196	3	2026-01-06 22:06:00	99	UPI	
36	TXN100036	321	16	2026-01-25 16:31:00	49	UPI	
44	TXN100044	309	15	2026-01-24 23:34:00	99	Netbanking	
56	TXN100056	132	24	2026-01-07 14:44:00	299	Wallet	
74	TXN100074	244	32	2026-01-18 07:44:00	149	UPI	
92	TXN100092	157	37	2026-01-28 18:27:00	499	Wallet	
130	TXN100130	211	3	2026-01-13 07:54:00	149	UPI	cl
134	TXN100134	74	27	2026-01-17 14:00:00	799	UPI	
188	TXN100188	32	31	2026-01-15 20:01:00	299	UPI	
212	TXN100212	167	37	2026-01-06 11:19:00	99	UPI	
250	TXN100250	130	39	2026-01-22 12:18:00	2999	Wallet	
259	TXN100259	317	9	2026-01-22 10:56:00	499	Wallet	
288	TXN100288	171	9	2026-01-19 23:48:00	299	UPI	
313	TXN100313	252	13	2026-01-10 22:58:00	99	Wallet	
324	TXN100324	177	9	2026-01-15 21:33:00	99	UPI	
330	TXN100330	329	11	2026-01-18 02:32:00	499	UPI	
351	TXN100351	186	36	2026-01-19 17:12:00	4999	UPI	
360	TXN100360	251	37	2026-01-17 16:34:00	49	UPI	
379	TXN100379	227	17	2026-01-03 11:59:00	299	Wallet	
390	TXN100390	149	3	2026-01-16 05:59:00	4999	Wallet	
395	TXN100395	173	19	2026-01-06 00:47:00	499	UPI	
471	TXN100471	283	31	2026-01-13 12:24:00	299	Netbanking	
475	TXN100475	291	31	2026-01-29 22:56:00	49	Netbanking	